BIRD CALL CLASSIFICATION USING NEURAL NETWORKS

Abstract

Convolutional neural networks (CNNs) are used to classify bird calls from 12 bird species commonly found in the Seattle area. The audio recordings are represented as mel spectrograms, which allow the model to learn frequency and timing patterns from the sounds.

The project includes both a binary classification task and a multiclass classification task. First, a model is trained to distinguish between two bird species. Then, the approach is expanded to classify all 12 species. Finally, the trained model is tested on three unlabeled audio recordings to evaluate how well it performs on unseen data.

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1.0 INTRODUCTION

Bird species identification from audio is an important problem in ecology and wildlife monitoring. Traditionally, this task has been performed manually by expert ornithologists who listen to bird calls and songs to identify species. However, manual identification is time consuming, requires specialized expertise, and is difficult to scale for large audio datasets collected in the field. Automated classification using machine learning provides a scalable alternative that can rapidly process large amounts of audio data while maintaining high accuracy.

The dataset used in this project comes from Xeno-Canto, a crowd-sourced archive of bird sound recordings collected from contributors around the world. The project focused on 12 bird species commonly found in the Seattle area, with 128 spectrogram samples per species for a total of 1,536 samples. Audio recordings were converted into mel spectrograms to prepare them for machine learning analysis. Spectrograms represent sound as a two-dimensional image showing frequency changes over time, making them well suited for image-based deep learning methods such as convolutional neural networks.

This project involved building multiple deep learning models for bird sound classification. First, a binary classifier was created to distinguish between two bird species. Next, a larger 12-class classifier was developed to identify all species included in the dataset. Multiple convolutional neural network architectures and hyperparameter settings were compared to evaluate performance improvements. The models were implemented using PyTorch, a deep learning framework that provides fine-grained control over model architecture and training. Finally, the trained models were applied to three unlabeled mystery recordings in order to predict and analyze the bird species present in each recording.

The trained models achieved high classification accuracy on the curated dataset, demonstrating strong performance in recognizing bird vocalizations. The mystery recordings were successfully classified and analyzed, and analysis of the recordings suggested that at least one clip likely contains calls from more than one species simultaneously. Overall, the results demonstrate that convolutional neural networks are highly effective for automated audio classification tasks and show strong potential for scalable wildlife monitoring applications.

2.0 THEORETICAL BACKGROUND

2.1 Spectrograms and the Mel Scale

A spectrogram is a visual representation of sound showing how frequencies change over time, with the horizontal axis representing time, the vertical axis representing frequency, and color intensity representing sound strength. In this project, bird vocalizations were converted into mel spectrograms using the mel scale, which better matches human auditory perception by emphasizing lower frequencies. Each spectrogram was standardized to 517 time steps by 128 mel frequency bins, representing a 3-second audio window sampled at 22,050 Hz, allowing spectrograms to be treated as images suitable for convolutional neural networks.

2.2 Neural Networks and Training

A neural network consists of interconnected neurons, each of which multiplies inputs by learned weights, adds a bias, and passes the result through an activation function. The Rectified Linear Unit (ReLU) is commonly used because it reduces vanishing gradient problems and enables efficient learning of complex patterns. As described in Chapter 10 of James et al. (2021), networks learn by adjusting weights to minimize a loss function through backpropagation, which calculates how much each weight contributed to the prediction error and updates them using gradient descent. The learning rate controls the size of each update step — too high causes instability and overshooting, too low causes slow convergence. In this project, a learning rate of 0.01 caused complete training failure, confirming its critical role in optimization (James et al., 2021, Section 10.7)

2.3 Convolutional Neural Networks

Fully connected networks treat every input independently, ignoring spatial structure and requiring excessive parameters for image data. Convolutional neural networks (CNNs) address this by applying small filters across the input to detect local patterns such as edges, frequency transitions, and harmonic structures. MaxPooling reduces spatial dimensions while retaining important features, and Batch Normalization stabilizes training by normalizing layer outputs for more consistent distributions. Dropout randomly deactivates neurons during training, preventing overfitting by forcing the network to learn redundant representations — a value of 0.5 was used in this project (James et al., 2021, Section 10.7). CNNs are especially well suited to spectrogram data because bird calls contain meaningful local patterns in both time and frequency dimensions (James et al., 2021, Section 10.3).

2.4 Loss Functions and Evaluation Metrics

A loss function measures prediction error and guides training. Binary Cross Entropy (BCE) is used for binary classification, comparing a single sigmoid probability to a binary label. Cross Entropy loss is used for multiclass classification, measuring confidence in the correct class across all outputs. A loss of 0.693 observed in the failed high learning rate experiment equals ln(2), indicating random guessing with no meaningful learning. Model performance is evaluated using precision, recall, and F1 score — precision measures correct positive predictions, recall measures identified actual positives, and F1 combines both as their harmonic mean. A confusion matrix visualizes correct and incorrect predictions across all classes, providing more insight than accuracy alone.

3.0 METHODOLOGY

3.1 Data Source and Description

The dataset used in this project was sourced from Xeno-Canto via the Kaggle birdcall competition. It contains recordings of 12 Seattle-area bird species, with 128 samples per species, resulting in a total of 1536 audio samples. The data was provided in HDF5 format, which allows efficient storage and retrieval of large numerical datasets. It was loaded into Python using the h5py library. Each audio sample was already preprocessed into spectrogram form using 3-second audio windows at a sample rate of 22050 Hz, ensuring consistency in time–frequency representation across the dataset.

3.2 Preprocessing
Although spectrograms were precomputed, frequency dimensions varied across species. All spectrograms were resized to a uniform shape of (517, 128) using PIL and normalized to range 0 to 1. A channel dimension was added for PyTorch CNN input format. The dataset was split 80/20 using stratified sampling, producing 1228 training and 308 test samples. PyTorch DataLoaders were created with batch size 32, with shuffling enabled for training to improve generalization and disabled for evaluation to ensure consistent results.

3.3 Binary Model
For the binary classification task, two species—American Crow and American Robin—were selected because they are acoustically distinct and show visually different frequency patterns in their spectrograms, making them a strong and interpretable baseline for the binary task. The rchitecture consisted of three convolutional layers with ReLU activation, followed by MaxPooling, Adaptive Average Pooling, a Dropout layer with a rate of 0.5, and a sigmoid output layer for binary prediction. The model was trained using BCELoss with the Adam optimizer and a learning rate of 0.001 for 20 epochs. Three configurations were compared in this experiment: the original model, a high learning rate version, and a simpler architecture to evaluate the impact of model complexity and optimization settings.

3.4 Multiclass Model
The multiclass model extended the task to classify all 12 bird species simultaneously. It used the same base convolutional architecture, with the addition of Batch Normalization layers and a fully connected layer with 256 neurons before the final output layer of 12 neurons. This model used CrossEntropyLoss and the Adam optimizer with a learning rate of 0.001. A learning rate scheduler was also applied, reducing the learning rate by half every 10 epochs to improve convergence. The model was trained for 30 epochs, and three different configurations were tested to compare performance under varying training conditions and architectural choices. The three configurations tested were the original model, a deeper four-layer network, and a lower learning rate variant of 0.0001.

3.5 Test MP3 Processing
To evaluate real-world performance, three unlabeled MP3 audio files were provided as test inputs. These files were loaded using librosa at a sample rate of 22050 Hz. A sliding window approach was applied using 3-second segments with a 1-second stride to capture multiple overlapping audio frames. Each segment was converted into a mel spectrogram, resized to match the training input shape, and normalized. The processed segments were passed through the trained 12-class model, and final predictions were determined using a majority vote across all windows. Additionally, probability distributions were plotted to analyze prediction confidence and assess model reliability across different audio segments.

4.0 RESULTS

4.1 Spectrograms overview

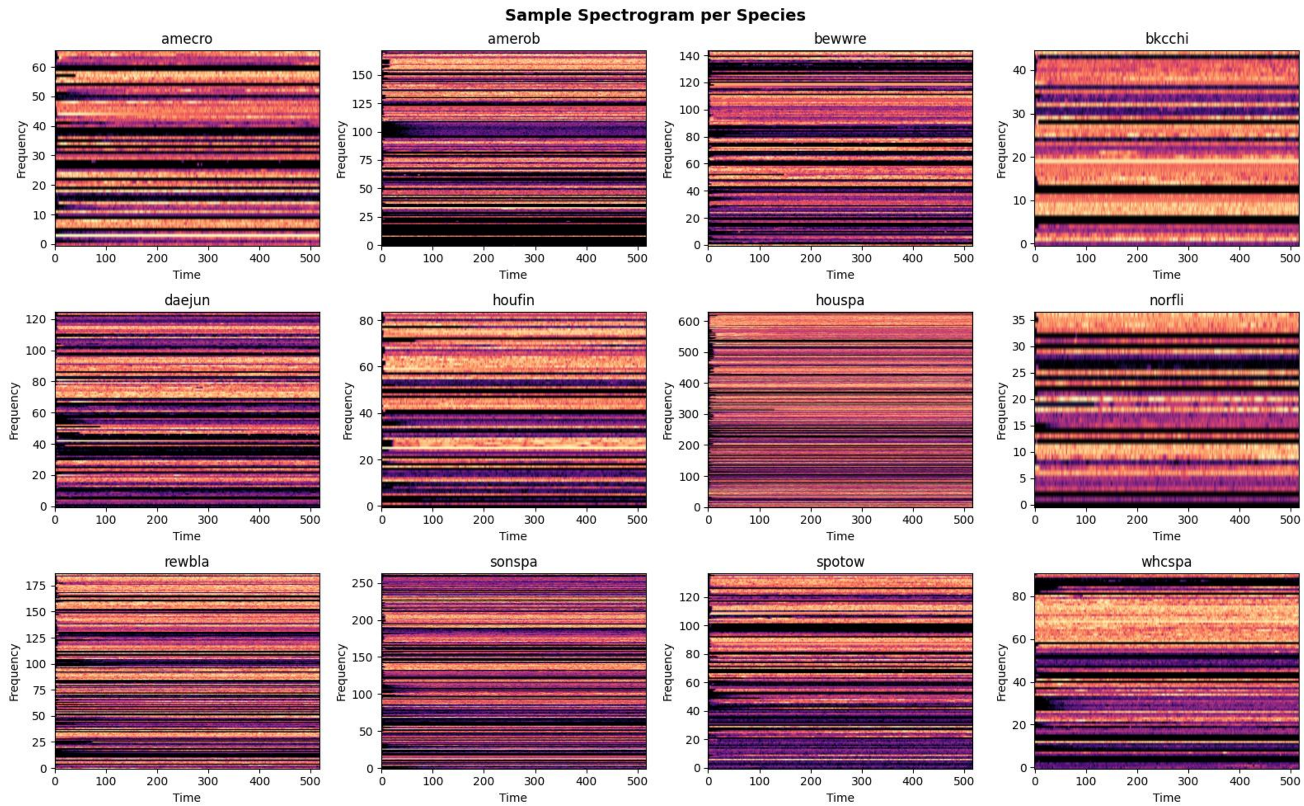

Figure 1: Sample Spectogram for all bird species

Figure 1 shows a sample mel spectrogram for each of the 12 bird species. Clear visual differences are visible across species in frequency range, pattern density and texture, with larger birds like the American Crow showing low-frequency energy and smaller songbirds showing high-frequency bursts. These differences suggest CNN-based classification should be effective for this dataset

4.2 Binary Training Curves

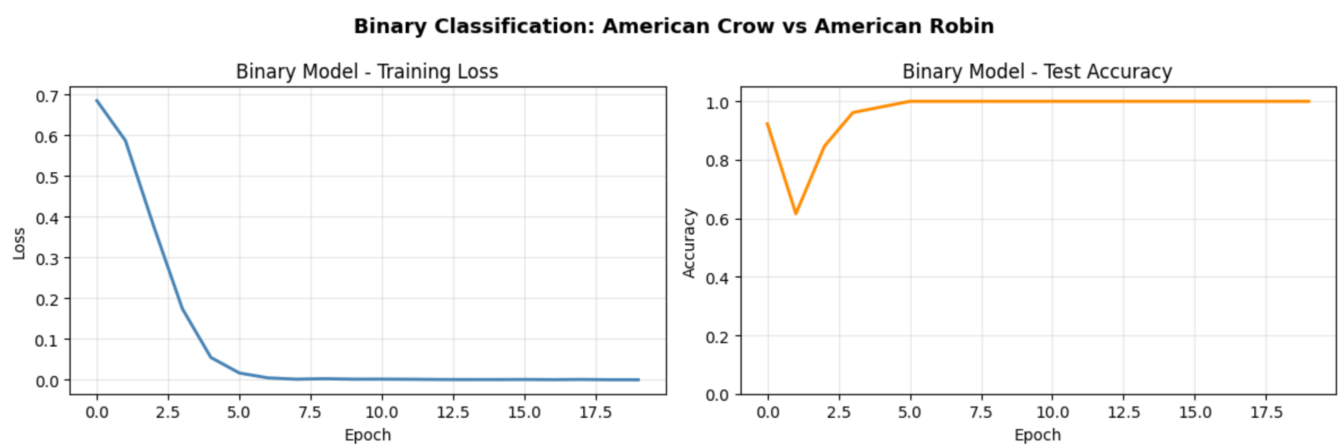

Figure 2: Binary Model- Training Loss and Test Accuracy

Figure 2 shows training loss and accuracy curves for the binary model. Loss drops sharply in early epochs and accuracy reaches 100% quickly, indicating fast and stable convergence. This confirms the two species are well separated in spectrogram feature space

4.3 Binary Comparison

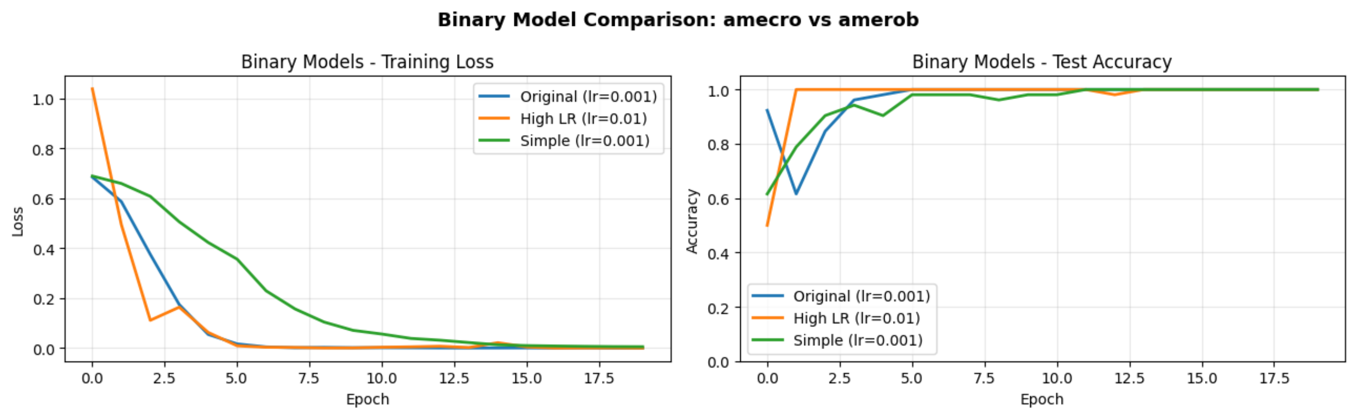

Figure 3: Binary Model- Comparison Plot

Figure 3 compares three binary model configurations. The high learning rate model fails completely at 50% accuracy while the simpler model reaches 98% but converges more slowly. The original model achieves the best balance of speed and accuracy.

4.4 Multiclass Comparison

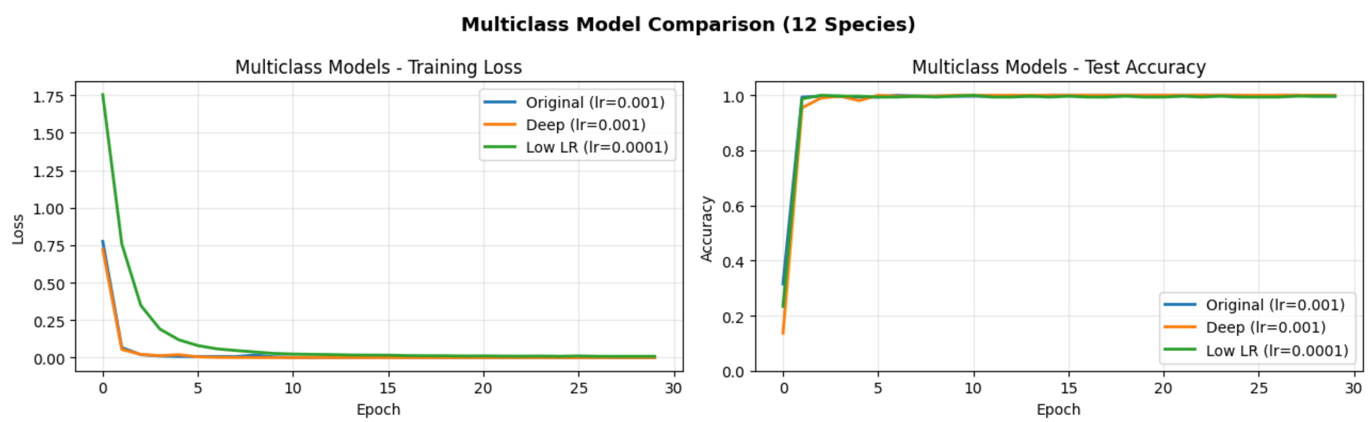

Figure 4: Multiclass Model Comparison- Training Loss and Test Accuracy

Figure 4 compares three multiclass configurations. The deeper model converges more slowly and takes over twice as long to train, while the low learning rate model shows gradual convergence with higher remaining loss. All three reach 100% accuracy, confirming the original model is the most efficient choice.

4.5 Confusion Matrix

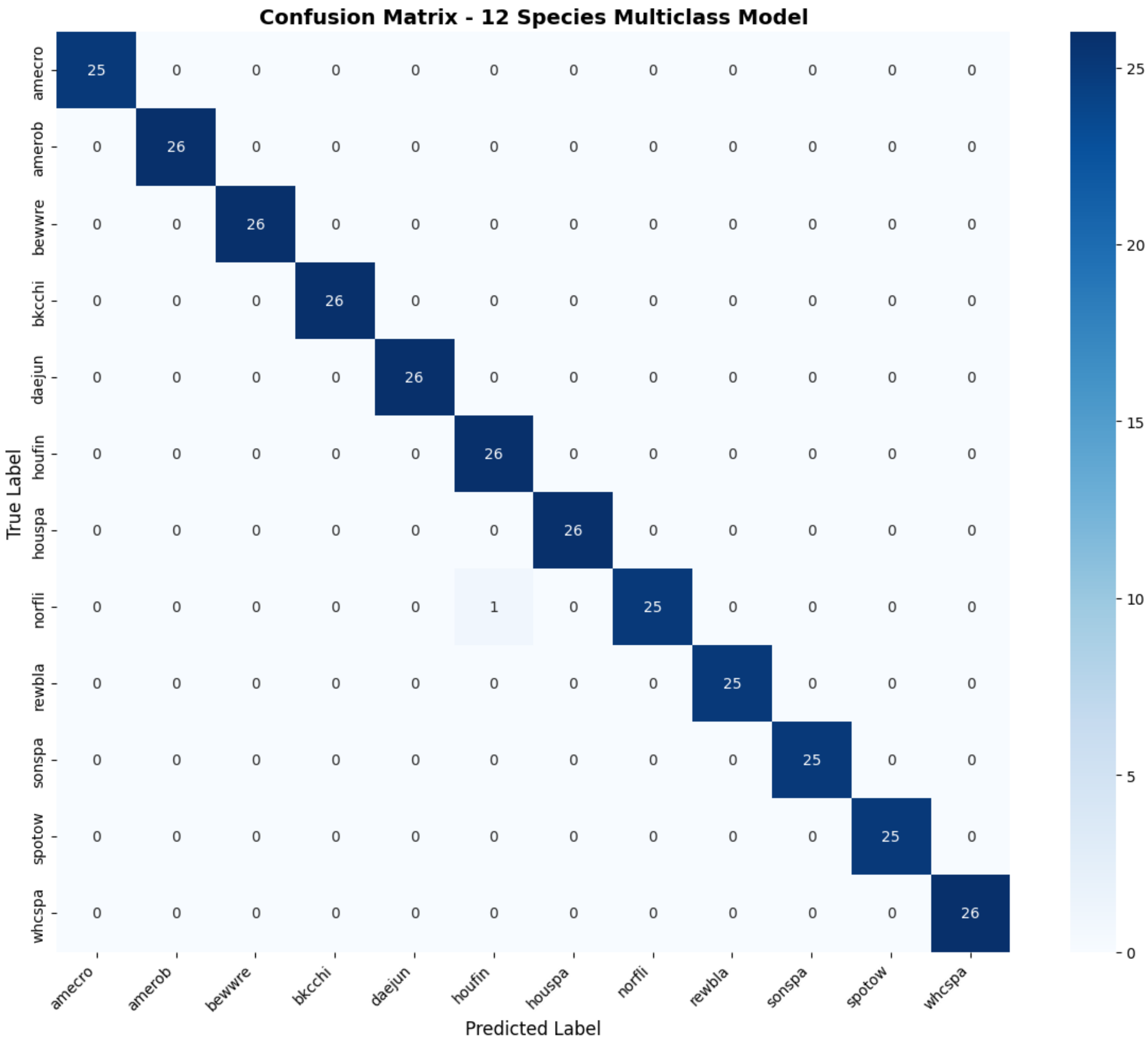

Figure 5: Multiclass Model Confusion Matrix

Figure 5 shows the confusion matrix for the 12-class model. All values fall on the diagonal with no off-diagonal entries, confirming perfect classification across all 308 test samples. While impressive, this likely reflects the curated nature of the dataset

4.6 Test predictions

Figure 6 shows probability distributions for the three test clips. All three predict bkcchi as the top species with houspa consistently second. test2 shows the lowest confidence at 50.6% with the most even split between two species, suggesting multiple birds may be present.

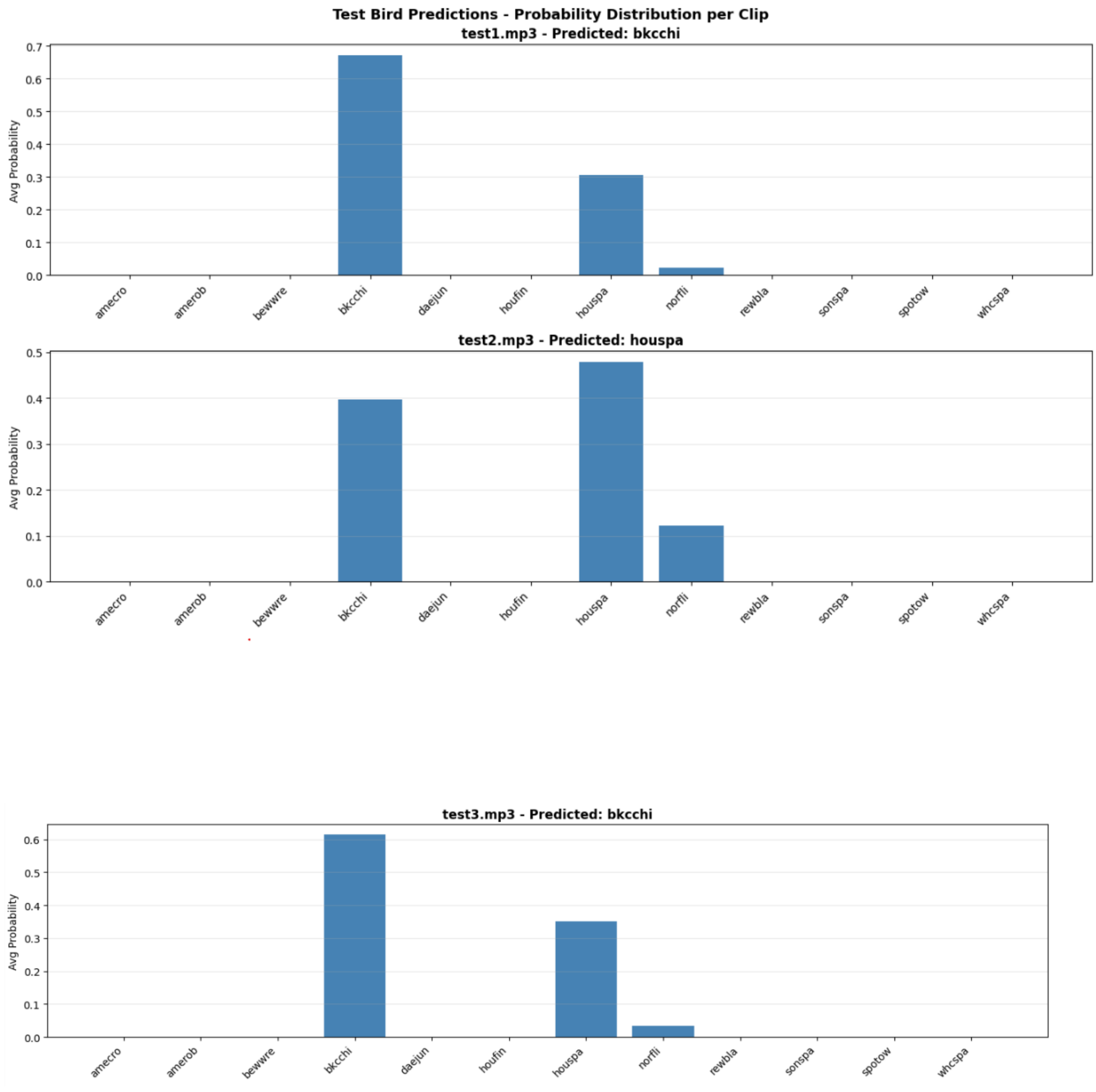

Figure 6: Test Predictions Summary

4.7 Test Spectograms

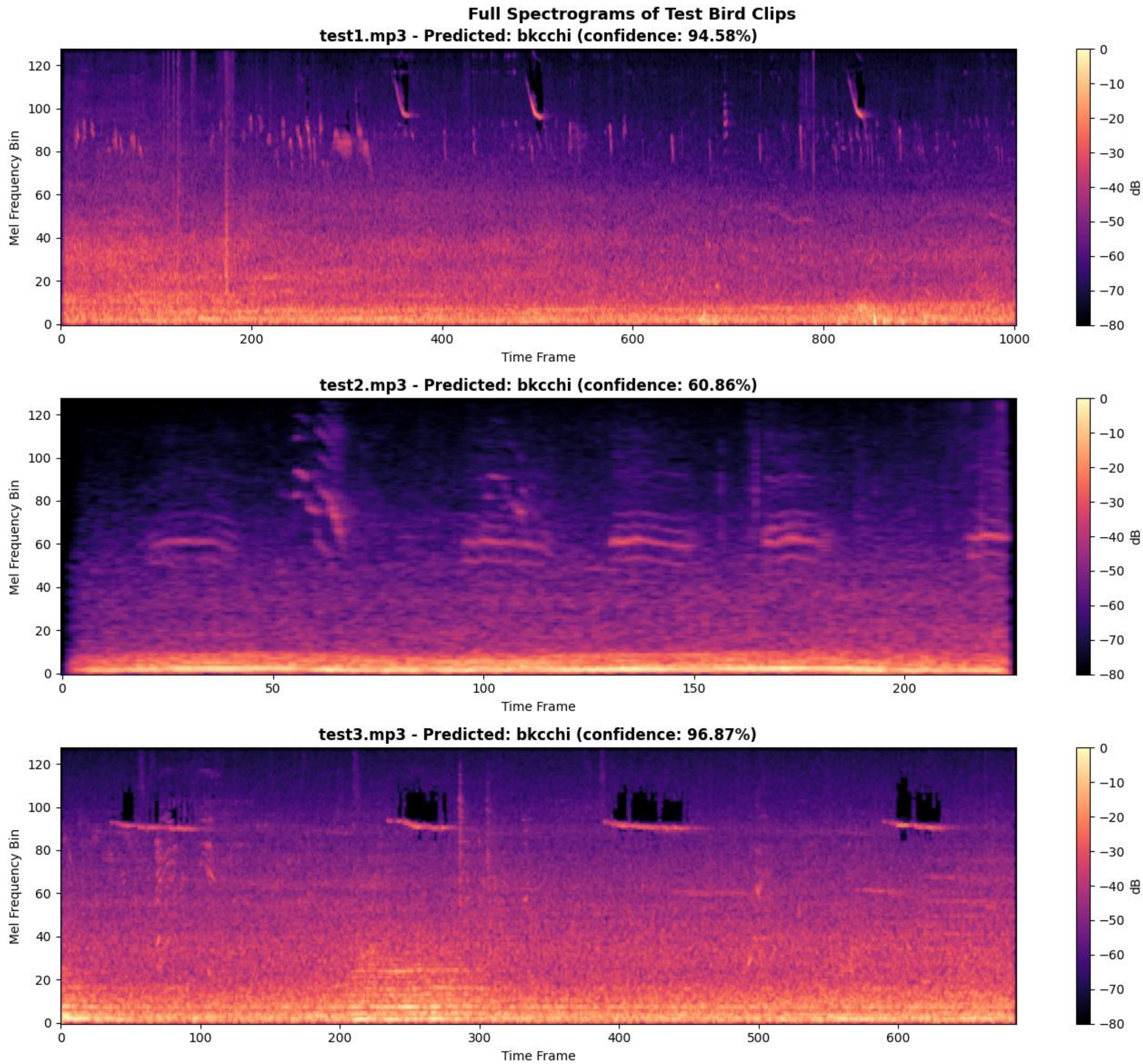

Figure 7: Spectrograms for the three test recordings used in the final evaluation.

Figure 7 displays full mel spectrograms of the three test recordings. test1 and test3 show structured concentrated frequency patterns while test2 shows more dispersed energy and less consistent structure. This visual evidence supports the conclusion that test2 likely contains overlapping bird calls.

5.0 DISCUSSION

5.1 Limitations

One key limitation of this project is the relatively limited scope of the dataset and computational training process. Although the dataset contains 12 species with a balanced number of samples per class, it is still relatively small compared to real-world bird audio distributions, where species are highly imbalanced and environmental noise is much more variable. Additionally, training time constrained experimentation with larger architectures or more extensive hyperparameter tuning. While the models converged quickly and achieved high accuracy, longer training with more varied augmentation strategies could potentially improve robustness and generalization. In terms of computational cost, the binary classification model trained in approximately 12 seconds over 20 epochs, while the multiclass model required approximately 56 seconds over 30 epochs, both using a T4 GPU on Google Colab.

5.2 Species difficulty and confusion

Although overall performance was perfect on the training data, the test clip analysis revealed recurring confusion between the Black-capped Chickadee (bkcchi) and House Sparrow (houspa). Both species produce short, high-pitched vocalizations with overlapping frequency ranges and similar harmonic structures. Listening to their calls on Xeno-Canto confirms these similarities — both are difficult to distinguish by ear in noisy conditions. Among all 12 species, sparrows such as sonspa, whcspa, and houspa are acoustically most similar, sharing comparable call durations and spectral energy distributions. Species like the American Crow and Northern Flicker are far easier to distinguish due to their distinctly different low-frequency or percussive call structure

5.3 Test Predictions Summary

Clip	Predicted Species	Confidence	Secondary Species	Multiple Birds?
test1.mp3	Black-capped Chickadee	71.6%	House Sparrow (25.7%)	Possibly
test2.mp3	Black-capped Chickadee	50.6%	House Sparrow (40.8%)	Likely yes
test3.mp3	Black-capped Chickadee	67.2%	House Sparrow (29.4%)	Possibly

5.4 Multiple birds analysis

The results also highlight the possibility of multiple-bird recordings influencing model uncertainty. In particular, the third test clip showed a more distributed probability spread across several classes and lower confidence in the top prediction. This aligns with the spectrogram analysis, which showed less structured frequency patterns and more variation over time. These characteristics are consistent with overlapping bird calls or background environmental noise. In real-world deployments, such mixed audio conditions would likely reduce classification accuracy unless additional source separation or segmentation techniques are applied.

5.5 Alternative models

Traditional methods such as support vector machines or random forests could be applied to flattened spectrogram features, but these treat each frequency bin independently and miss the spatial relationships that define bird call shapes. Neural networks make particular sense here because bird call classification is fundamentally a pattern recognition problem on 2D spectrogram images. CNNs automatically learn hierarchical spatial features — frequency bursts, harmonics, and temporal patterns — without manual feature engineering, making them the natural choice. Future work could explore ResNet architectures, attention-based models, or SpecAugment data augmentation to improve robustness on noisier recordings

6.0 CONCLUSION

Overall, this project demonstrates that convolutional neural networks are highly effective for bird species classification using spectrogram representations of audio. The models achieved near-perfect to perfect performance on both binary and multiclass tasks, showing that spectrograms contain strong discriminative features that CNNs can learn efficiently. The comparison of different learning rates and architectures highlighted the importance of stable optimization, with overly high learning rates causing complete training failure and simpler models reducing learning efficiency.The analysis of test MP3 files further showed that the model can generalize to unseen audio segments using a sliding window approach, producing consistent predictions across time. However, cases with lower confidence and more distributed probabilities suggest that real-world audio complexity, including overlapping species and environmental noise, remains a significant challenge. In conclusion, while the model performs exceptionally well on curated data, future improvements in dataset diversity, augmentation, and model architecture would be necessary for deployment in real-world bioacoustic monitoring systems.

7.0 REFERENCES

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